Yanis Bendehane

Engineering Student in Computer Science | Critical and Embedded Systems

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Summary

Computer science engineering student in an apprenticeship at Thales, working on mission-critical defense software. Background in embedded prototyping (Arduino, sensors), system modeling (SysML, UML) and industrial real-time data. Seeking a 17-week research internship abroad in embedded systems or IoT, mid-June to mid-October 2027.

Education

CESI School of Engineering, MEng in Computer Science – Aix-en-Provence, France Sept 2023 – Sept 2028

- French Diplôme d'Ingénieur (CTI-accredited, Master's level): integrated preparatory cycle 2023–2025, apprenticeship engineering cycle with Thales 2025–2028
- Engineering cycle: semester 5 fully validated with grade A in every teaching unit; operations research unit graded A

Lycée Jean Monnet, Bac in Mathematics and Physics-Chemistry, with honors (Mention Bien) – Vitrolles, France Sept 2020 – July 2023

Experience

Apprentice Software Engineer, Thales Services Numériques – Aix-en-Provence, France Oct 2025 – present

- Develop and test features of a large-scale, mission-critical defense software system in Java on Linux, within a 40-person engineering team spread over three sites
- Apply the reliability and quality standards of critical systems: strict configuration management (Subversion), testing and documentation

IT Intern, LyondellBasell – Berre-l'Étang, France Apr 2025 – June 2025

- Rebuilt a legacy C++ (MFC) program as a standalone C# Windows service that synchronizes tank logistics states between a real-time industrial database (AspenTech IP.21) and SQL Server over ODBC
- Added detailed logging of every update (state before/after, connections, errors) for a high-risk (Seveso) industrial site

Academic Projects

Embedded weather station for ships (group project, Embedded Systems module) Fall 2024

- Co-developed the Arduino Uno firmware (C++) of an autonomous station acquiring temperature, humidity, pressure, light, GPS position and real-time clock over I2C, SPI and UART, with CSV logging to an SD card
- Interrupt-driven operating modes with RGB LED status feedback, modular object-oriented code, designed first in UML/SysML

Secure safe prototype on Arduino (team of 4, Electronics module) Fall 2023

- Designed a multi-factor access logic (switch code + card) from truth tables and Karnaugh maps to a logic circuit, then prototyped it on Arduino

Game of Life in C++ (team of 3, Object-Oriented Programming module) Dec 2024

- Developed an object-oriented simulation with console and graphical modes (SFML), toroidal grid, predefined patterns and file loading

Leadership and Volunteering

Co-organizer, charity football tournament (CESI civic engineering project) Jan 2024 – June 2026

- Organized three editions with a team of five; proceeds donated to charities chosen by the winning teams (incl. Les Restos du Cœur, ShareTheMeal)

Skills

Programming: C, C++, Java, C#, Python

Embedded: Arduino (C++), I2C, SPI, UART, sensors, GPS, RTC, interrupts, digital logic, SysML and UML modeling

Systems: Linux, Windows services, real-time industrial data (AspenTech IP.21, ODBC), critical-systems quality practices

Tools: Arduino IDE, Git, Subversion, Visual Studio, IntelliJ IDEA

Languages: French (native), English (B2, TOEIC practice test 865/990), Spanish (school level), Arabic (basic)

Certification: ANSSI SecNumacadémie (French national cybersecurity MOOC)